

© Zahra Shakeri, Dalla Lana School of Public Health, University of Toronto



Institute of Health Policy, Management and Evaluation
UNIVERSITY OF TORONTO

Dalla Lana
School of Public Health

FUNCTIONS



In mathematics, a function is a relation between input x and output y sets, where each x maps to a unique y , denoted as $y = f(x)$.



This notation means that for a given ' x ' value, the function ' f ' will produce a ' y ' value.

Example– We want to explore the relationship between cigarette smoking and lung cancer.

Our function (f) connects the daily number of cigarettes smoked (input $\mathbf{x}$) to the likelihood of developing lung cancer (output $\mathbf{y}$).



FUNCTION

In mathematics, a function is a relation between input x and output y sets, where each x maps to a unique y , denoted as $y = f(x)$.

This notation means that for a given 'x' value, the function 'f' will produce a 'y' value.

Example— We want to explore the relationship between cigarette smoking and lung cancer.

Our function (f) connects the daily number of cigarettes smoked (input x) to the likelihood of developing lung cancer (output y).



FUNCTION

In mathematics, a function is a relation between input x and output y sets, where each x maps to a unique y , denoted as $y = f(x)$.

This notation means that for a given 'x' value, the function 'f' will produce a 'y' value.

We have conducted a study and found a linear relationship between the number of cigarettes smoked daily and the risk of developing lung cancer over a 40-year span.

- If a person does not smoke, the risk is **1%**.
- For every cigarette smoked daily, the risk increases by **0.5%**.

$$y = 0.01 + 0.005x$$

↓
Lung cancer risk
over 40 years

↓
The # of cigarettes
smoked daily

